

Continuing Professional Development Log

Name: Zheng Xu

Student ID: 11314389

Current and recent CPD activity:

CPD Activity Title	Description	Dates	CPD Hours
Project planning and CPD setup	Reviewed the EEN30330 handbook requirements for weekly project meetings, progress review meetings, project logbook evidence, self-directed learning and skill development. Set up the project plan, risk assessment tracking and a habit of recording supervisor actions and technical progress.	06 Oct-20 Oct 2025	6
MI-BCI literature review	Studied motor imagery BCI literature, CTNet/EEGTransformer approaches, EEGNet, CSP-based baselines and reinforcement-learning BCI work. This developed the background needed to justify the hybrid classifier and closed-loop controller choices in the report.	Oct-Nov 2025	16
EEG preprocessing with MNE-Python	Learned how to load PhysioNet and BCI Competition data, map labels, epoch motor-imagery trials, apply the 8-30 Hz bandpass filter, normalise data and maintain participant-wise train/test splits. This supported reproducible EEG classification experiments.	Nov-Dec 2025	18
PyTorch CTNet and benchmark evaluation	Implemented and adapted the CTNet/EEGTransformer model in PyTorch, including training loops, evaluation reports and saved checkpoints. Benchmarks were run on BCI IV-2a, BCI IV-2b and the larger PhysioNet motor-imagery dataset.	Dec 2025-Jan 2026	24
Transfer learning and channel reduction	Developed subject-specific fine-tuning, filter/ICA comparisons and channel-reduction experiments. The work linked technical decisions to deployment constraints, including an OpenBCI-oriented 8-channel setup and the observed importance of motor-cortex channels such as C3.	Feb-Mar 2026	28
RL control and simulation demos	Studied DQN control, reward shaping, Gym/PyBullet simulation and trajectory diagnostics. I reviewed the three available demonstration media: gym_arm.gif as a 500x500, 10-frame, 5 s simulation demo, and rl_arm.gif in both 03_Experiments and 05_Documentation as identical 500x500, 20-frame, 10 s RL trajectory animations. This improved how I explain controller behaviour in the final demonstration.	Mar-Apr 2026	22
BrainFlow, OpenBCI and SO-101 integration	Learned the BrainFlow acquisition workflow, synthetic-board testing, online buffer handling, classifier-to-policy action mapping and serial control of the SO-101 arm. Safety-related learning included workspace limits, velocity limiting, clear operating zones and immediate power-off access.	Apr 2026	16
Report, repository and presentation preparation	Prepared final-report evidence, repository structure, README material, figures and appendices. This activity focused on academic integrity, clear referencing, traceable software artefacts and a concise technical narrative for the final demonstration/presentation.	Apr-May 2026	14

Planned and Future CPD activity:

CPD Activity Title	Description	Skills addressed	Dates
Final demonstration rehearsal	Practise the 15-minute final presentation, rehearse responses to questions about classifier accuracy, simulation-only validation, ethical constraints and the RL demonstration media.	Technical communication; evidence-based explanation; time management.	05 May-15 May 2026
Repository release and reproducibility check	Clean the public repository evidence so that a technically competent reader can inspect the code structure, run key scripts and understand which artefacts are authored work versus third-party reference material.	Version control; reproducible research; professional software documentation.	May 2026
Ethically approved OpenBCI user study	Prepare a live-user validation plan with participant information, consent, risk controls and a small 8-channel calibration protocol before collecting any new human EEG data.	Research ethics; participant protocol design; EEG acquisition practice.	Summer-Autumn 2026
Real-time 8-channel inference path	Build and test a native low-channel online decoder using the exact OpenBCI channel order, sampling rate, filtering and normalisation expected during deployment. Compare shorter decision windows to reduce command latency.	Online signal processing; model adaptation; latency analysis.	Summer 2026
Extended SO-101 closed-loop manipulation	Extend the present planar reaching controller toward richer SO-101 tasks only after validating live decoding and emergency-stop controls. Use the current 4-direction policy as the safety baseline.	Robotics control; reinforcement-learning generalisation; safety validation.	Future MEng/research work